3.2

✓ 2 (2)

$$y = \sqrt[3]{\frac{1+x^2}{1-x^2}}$$

$$y^3 = \frac{1+x^2}{1-x^2}$$

$$3y^2 \cdot y' = \frac{2x(1-x^2) + 2x(1+x^2)}{(1-x^2)^2} = \frac{4x}{(1-x^2)^2}$$

$$\therefore y' = \frac{4x}{(1-x^2)^2} \cdot \frac{1}{3y^2} = \frac{4}{3} \cdot \frac{x}{(1-x^2)^2} \cdot \left(\frac{1+x^2}{1-x^2}\right)^{-\frac{2}{3}}$$

✓ 3. $\frac{d}{dx}\{(3x-1)^8 f(2x+3)\}$

$$= 8(3x-1)^7 \cdot 3 \cdot f(2x+3) + (3x-1)^8 \cdot f'(2x+3) \cdot 2$$

$$= 24(3x-1)^7 \cdot f(2x+3) + 2(3x-1)^8 \cdot f'(2x+3)$$

✓ 6. $f(x) = \frac{2+x}{2-x}$ $A(a, f(a))$

$$f'(a) \circ \left(-\frac{1}{4}\right) = -1$$

$$f'(x) = \frac{(2-x) + 2+x}{(2-x)^2}$$

$$\frac{4}{(2-a)^2} \cdot \left(-\frac{1}{4}\right) = -1$$

$$\therefore (2-a)^2 = 1$$

$a = 1$ or 3

답: $(1, 3)$, $(3, -5)$

10 (3) $0 = x^2 - 6x + y^2 - 2y - 15$ $P(-1, 4)$

양변 미분

$$2x - 6 + 2y \cdot \frac{dy}{dx} - 2 \cdot \frac{dy}{dx} = 0$$

$P(-1, 4)$ 대입

$$-2 - 6 + 8 \cdot \frac{dy}{dx} - 2 \cdot \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{4}{3} \text{ 이다.}$$